

Software Requirements Specification (SRS)

XRayVision AI — Web-Based AI Chatbot for Automated X-Ray Fracture Detection

Field	Value
Document	Software Requirements Specification
Version	1.0
Standard	IEEE Std 830-1998 (adapted)
Project	XRayVision AI — Full-Stack Medical Diagnostic Platform
Institution	Minhaj University Lahore — School of Software Engineering
Programme	BSSE, 8th Semester — Final Year Project
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Table of Contents

1. [Introduction](#)
 2. [Overall Description](#)
 3. [External Interface Requirements](#)
 4. [Functional Requirements](#)
 5. [Use Case Model](#)
 6. [Non-Functional Requirements](#)
 7. [Other Requirements](#)
 8. [Appendices](#)
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1. Introduction

1.1 Purpose

This document specifies the complete software requirements for **XR**ay**V**ision **AI**, a web-based medical diagnostic platform that analyses medical images using a multi-model deep-learning ensemble and produces a structured clinical report through an LLM synthesis layer.

The intended audience is:

- The project supervisor and the FYP evaluation committee
- The development team (for implementation and verification)
- Future maintainers who extend the system

This SRS defines **what** the system must do. The **how** is specified in the companion [Software Design Document](#).

1.2 Scope

Product name: XR

ay**V**ision **AI**

XR

ay**V**ision **AI** allows an authenticated user to upload a chest X-ray, a bone X-ray, or an external wound photograph and receive, within seconds, an AI-generated diagnostic report containing named findings, confidence scores, ICD-10 codes, bounding-box localisation, an urgency rating, recommended actions and a specialist referral suggestion.

In scope

#	Capability
1	User registration, login, password reset, profile and settings management
2	Chest X-ray pathology detection (18 classes)
3	Bone fracture localisation with bounding boxes + image-level fracture screening
4	External wound classification
5	LLM synthesis of model outputs into a readable clinical report
6	Permanent scan history with PDF and JSON export
7	Bilingual (English / Urdu) health chatbot with voice input and session persistence
8	Personalised 7-day diet plan generation with medical guardrails
9	GPS-based nearby clinic and hospital locator via OpenStreetMap
10	Dashboard analytics over the user's own scan history

Out of scope

- Clinical or diagnostic certification of any kind — the system is an educational aid only
- PACS / HL7 / FHIR hospital-system integration
- Multi-tenant hospital administration, billing, or appointment booking
- Real-time collaboration or multi-radiologist review workflows
- Training of new models from user-uploaded data

1.3 Definitions, Acronyms and Abbreviations

Term	Definition
AI	Artificial Intelligence
API	Application Programming Interface
CLAHE	Contrast Limited Adaptive Histogram Equalisation — contrast enhancement applied during preprocessing
CNN	Convolutional Neural Network
DenseNet121	A 121-layer densely-connected CNN; here pretrained by TorchXRayVision on 700,000+ clinical chest X-rays
DICOM	Digital Imaging and Communications in Medicine — the standard medical imaging file format (.dcm)
FYP	Final Year Project
ICD-10	International Classification of Diseases, 10th revision — standard diagnostic coding system
JWT	JSON Web Token — the stateless authentication token format used by the API
LLM	Large Language Model
Overpass API	The query API of the OpenStreetMap project, used to look up healthcare facilities
RLS	Row Level Security — PostgreSQL feature restricting row access per authenticated user
SPA	Single Page Application
SSR	Server Side Rendering
ViT	Vision Transformer — a transformer-based image classification architecture
YOLOv8	"You Only Look Once" v8 — a single-pass object detection model used for fracture localisation

1.4 References

#	Reference
R1	IEEE Std 830-1998, <i>Recommended Practice for Software Requirements Specifications</i>
R2	Cohen J.P. et al., <i>TorchXRayVision: A library of chest X-ray datasets and models</i> , https://github.com/mlmed/torchxrayvision
R3	Ultralytics YOLOv8 Documentation, https://docs.ultralytics.com
R4	Dosovitskiy A. et al., <i>An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale</i> , ICLR 2021
R5	FastAPI Documentation, https://fastapi.tiangolo.com
R6	TanStack Start / Router / Query Documentation, https://tanstack.com

#	Reference
R7	Supabase Documentation, https://supabase.com/docs
R8	OpenRouter API Documentation, https://openrouter.ai/docs
R9	OpenStreetMap Overpass API, https://wiki.openstreetmap.org/wiki/Overpass_API
R10	WHO, <i>International Classification of Diseases 10th Revision (ICD-10)</i>
R11	XRayVision AI — Software Design Document, docs/02-SDD.md

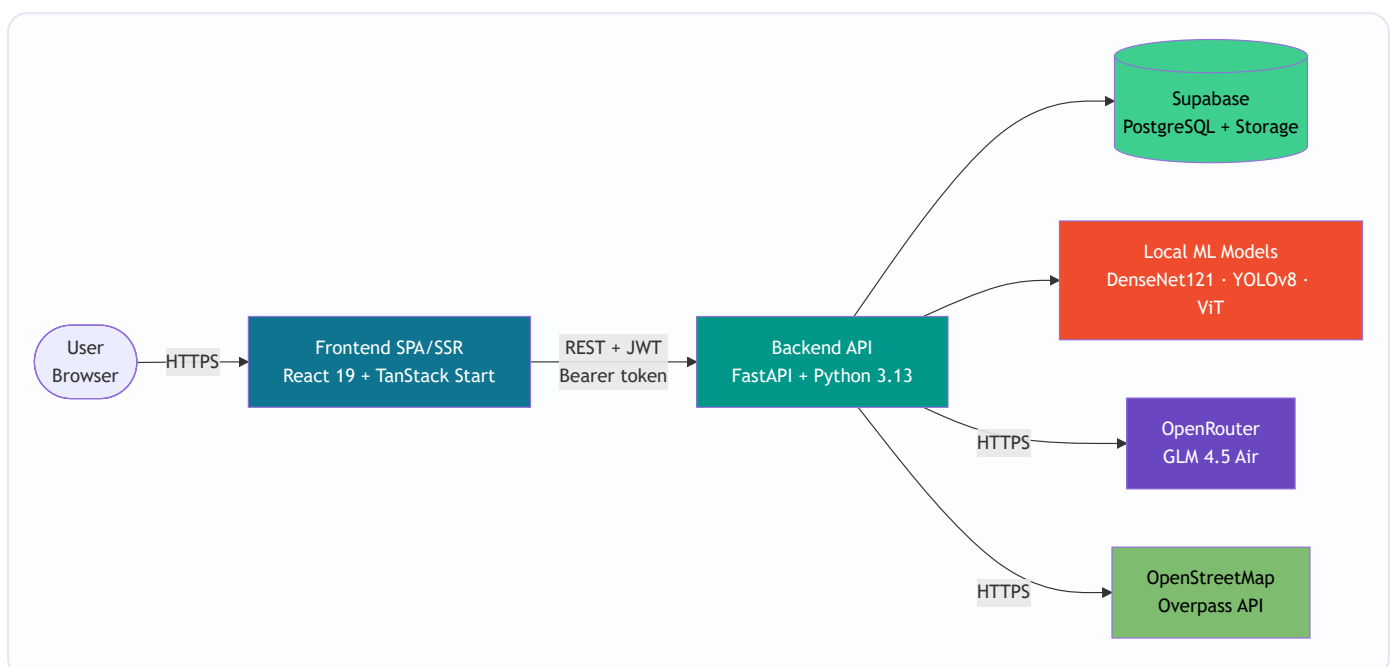
1.5 Overview of the Document

Section 2 describes the product at a high level — its context, its users and its constraints. Section 3 defines the external interfaces. Section 4 enumerates every functional requirement with a unique identifier. Section 5 presents the use case model. Section 6 states the measurable non-functional requirements. Sections 7 and 8 cover legal constraints and supporting appendices.

2. Overall Description

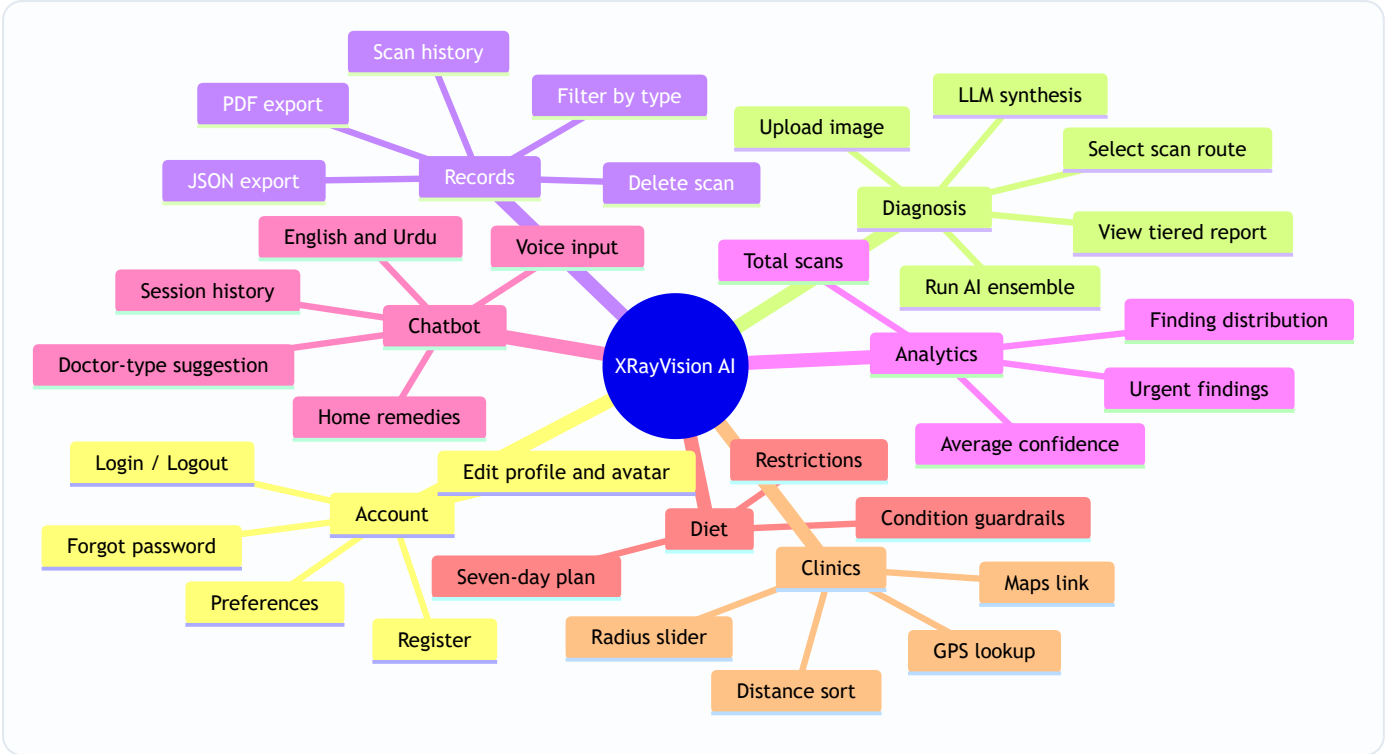
2.1 Product Perspective

XRayVision AI is a **new, self-contained web product**. It is not a replacement for, nor a component of, an existing system. It is built as three independently deployable tiers communicating over HTTPS:



The system depends on four **external services** it does not own: Supabase (database, auth backing store and object storage), OpenRouter (LLM inference), Hugging Face Hub (model weight distribution) and OpenStreetMap (facility lookup).

2.2 Product Functions



2.3 User Classes and Characteristics

User class	Technical skill	Frequency of use	Primary functions used	Priority
Medical Student (default role)	Moderate — familiar with web apps, learning radiology	Daily during study periods	Analysis, history, chatbot	High
Radiologist / Doctor	Moderate — domain expert, not a developer	Occasional, for preliminary screening	Analysis, PDF export, clinics	High
Healthcare Professional (nurse, dietitian)	Basic	Weekly	Chatbot, diet planner, clinics	Medium
Evaluator / Examiner	Any	One-off	Demo account, all features	Medium
System Administrator	High — developer	Setup and maintenance only	Deployment, environment configuration, Supabase console	Low

Note: The system does not implement role-based access control. `role` is a descriptive profile attribute (default `"Medical Student"`); all authenticated users have identical permissions over their own data.

2.4 Operating Environment

Client side

Item	Requirement
Browser	Chrome 120+, Edge 120+, Firefox 120+, Safari 17+
JavaScript	Must be enabled
Screen	Minimum 360 px width (responsive down to mobile)
Permissions	Geolocation required only for the Clinic Locator; microphone required only for chatbot voice input
Network	Broadband recommended (image upload up to 20 MB)

Server side

Item	Requirement
Backend runtime	Python 3.11+ (3.13 in production)
Backend framework	FastAPI 0.115, served by Uvicorn
Container	Docker, listening on port 7860 (Hugging Face Spaces convention)
Memory	≥ 2 GB RAM recommended for model preloading; <code>DISABLE_PRELOAD=true</code> supported on ≤ 512 MB free tiers
Frontend runtime	Node.js 18+ for build; served as a static + SSR bundle
Database	Supabase-managed PostgreSQL 15+

2.5 Design and Implementation Constraints

ID	Constraint
C-1	The system must remain deployable entirely on free-tier cloud services (Vercel, Hugging Face Spaces, Supabase free plan, OpenRouter free models). This is a hard FYP budget constraint.
C-2	The LLM must be <code>z-ai/glm-4.5-air:free</code> via OpenRouter. No paid inference provider may be required.
C-3	Model weights must come from public sources (TorchXRayVision, Hugging Face Hub) or be bundled in the repository (<code>backend/models/fracture_yolov8.pt</code>).
C-4	All AI outputs must carry an explicit "educational use only — not a medical device" disclaimer.
C-5	The backend must start and accept requests before model loading completes, so free-tier health checks do not time out.
C-6	No patient-identifying information may be requested or stored. Uploaded images are keyed only by an opaque user UUID.
C-7	Authentication must be stateless (JWT), because the backend host may cold-start or scale to zero.

ID	Constraint
C-8	Rate limiting must be applied to every LLM-backed endpoint to stay within free-tier quotas.
C-9	All model inference must run on CPU ; no GPU may be assumed.

2.6 Assumptions and Dependencies

Assumptions

- **A-1** Users upload genuine medical images. The system validates format and quality but cannot verify clinical provenance.
- **A-2** Users understand the output is advisory and not a diagnosis.
- **A-3** The uploading user has the right to upload the image.

Dependencies

Dependency	Failure impact	Mitigation implemented
Supabase availability	Scans cannot be saved or listed	Image upload failure is non-blocking — analysis still returns to the user
OpenRouter availability	No synthesis paragraph	Hard-coded fallback synthesis returned with <code>urgency: "medium"</code>
Hugging Face Hub	ViT / fracture classifier cannot download	Per-model try/except; other models still run, errors recorded in <code>model_results.model_errors</code>
YOLO weight file present	Fracture localisation unavailable	Image-level fracture screening classifier still runs
Overpass API	Clinic search unavailable	Three mirror endpoints are tried in sequence before failing

3. External Interface Requirements

3.1 User Interfaces

The frontend uses two layout shells:

Shell	Used by	Description
AuthShell	Login, Register, Forgot Password	Split layout — marketing panel plus a centred credential card
AppShell	All authenticated pages	Collapsible left sidebar + top header with page title, search and user menu

Sidebar navigation (fixed order): Dashboard · New Analysis · Health Chat · Diet Planner · Clinics · History · Profile · Settings.

ID	Requirement
UI-1	All pages shall be responsive from 360 px to 1920 px width.
UI-2	Light, dark and system theme modes shall be selectable in Settings and persisted per user.
UI-3	Every long-running action (analysis, chat reply, diet generation) shall display a visible loading state.
UI-4	Urgency shall be communicated by both colour and text label, so the UI never relies on colour alone.
UI-5	Every error returned by the API shall be surfaced as readable text, never as a raw stack trace.
UI-6	An "Educational use only" badge shall be permanently visible in the application shell.
UI-7	Findings shall be presented in three confidence tiers so the user can immediately judge reliability (see FR-028a).

3.2 Hardware Interfaces

ID	Interface	Purpose
HW-1	Browser Geolocation API	Supplies latitude/longitude to the Clinic Locator. If permission is denied the feature degrades gracefully with an explanatory message.
HW-2	Browser SpeechRecognition API (microphone)	Optional voice input for the health chatbot, using locale <code>en-US</code> or <code>ur-PK</code> . Text input remains fully available.

No GPU is required — all models run on CPU inference.

3.3 Software Interfaces

Interface	Type	Direction	Data exchanged
Supabase PostgreSQL	Supabase Python client	Backend $\rightleftharpoons$ DB	Profile, scan, chat session and chat message rows
Supabase Storage	Supabase Python client	Backend $\rightarrow$ Storage	Uploaded images (<code>xray-images</code> bucket), avatars (<code>avatars</code> bucket); returns a public URL
Supabase Auth	Supabase Python client	Backend $\rightleftharpoons$ Auth	User creation, credential verification, password-reset email
OpenRouter Chat Completions	HTTPS REST	Backend $\rightarrow$ OpenRouter	Structured clinical prompt; returns synthesis JSON. Model <code>z-ai/glm-4.5-air:free</code> , timeout 60 s
Hugging Face Hub	HTTPS	Backend $\rightarrow$ HF	Downloads ViT wound classifier and fracture screening classifier weights on first use
OpenStreetMap Overpass	HTTPS	Backend $\rightarrow$ Overpass	Healthcare facility query within a radius; three mirrors tried in order
TorchXRayVision	Python library	In-process	Supplies pretrained DenseNet121 weights
Ultralytics	Python library	In-process	Runs YOLOv8 inference on the bundled <code>.pt</code> weights

3.4 Communications Interfaces

ID	Requirement
CM-1	All client-server communication shall use HTTPS in production.
CM-2	The API shall be RESTful, accepting and returning <code>application/json</code> , except <code>POST /analyze (multipart/form-data)</code> and <code>GET /scans/{id}/report.pdf (application/pdf)</code> .
CM-3	Authenticated requests shall carry the token in an <code>Authorization: Bearer <jwt></code> header.
CM-4	CORS shall permit only the configured frontend origin(s) in production; localhost origins are permitted only when <code>FRONTEND_URL</code> is itself a localhost URL.
CM-5	Allowed methods shall be <code>GET, POST, PUT, PATCH, DELETE, OPTIONS</code> ; allowed headers <code>Authorization, Content-Type, Accept</code> .

4. Functional Requirements

Priority: **M** = Must have, **S** = Should have, **C** = Could have.

4.1 Module: Authentication & Account (FR-001 – FR-010a)

ID	Requirement	Priority
FR-001	The system shall allow a new user to register with full name, email, password and an optional role.	M
FR-002	The system shall reject a password shorter than 8 characters at both client and server.	M
FR-003	The system shall reject registration with an existing email, returning a readable error.	M
FR-004	On successful registration the system shall create a <code>profiles</code> row via a database trigger and return a JWT, signing the user in immediately.	M
FR-005	The system shall authenticate a returning user by email and password and return a JWT plus the user profile.	M
FR-006	The system shall issue JWTs signed with HS256 expiring after 24 hours (configurable via <code>JWT_EXPIRY_HOURS</code>).	M
FR-007	The system shall reject any request to a protected endpoint carrying a missing, malformed or expired token, with HTTP 401 .	M
FR-008	The system shall provide a "forgot password" flow that triggers a Supabase password-reset email.	S
FR-009	The system shall allow a signed-in user to update their full name, role and avatar image.	S
FR-010	The system shall allow logout by clearing the locally stored token and user object; no server call is required.	M
FR-010a	The login page shall provide a one-click demo account button so evaluators can explore the system without registering.	S

4.2 Module: Image Analysis (FR-011 – FR-025a)

ID	Requirement	Priority
FR-011	The system shall let the user select exactly one analysis route: <code>chest</code> , <code>fracture</code> , or <code>wound</code> .	M
FR-012	The system shall accept image uploads in JPEG, PNG and DICOM (.dcm) formats.	M
FR-013	The system shall reject a file smaller than 100 KB with HTTP 422 and an explanatory message.	M
FR-014	The system shall reject a file larger than 20 MB with HTTP 422 and an explanatory message.	M
FR-015	The system shall reject a non-DICOM image whose width or height is below 200 px with HTTP 422.	M
FR-016	The system shall reject an empty upload with HTTP 400.	M
FR-017	The system shall reject an invalid <code>scan_type</code> value with HTTP 400.	M
FR-018	The system shall allow an optional session/patient label and optional free-text clinical notes per analysis.	S
FR-019	The system shall apply CLAHE contrast enhancement and model-specific resizing/normalisation before inference.	M
FR-020	The system shall route the image correctly: <code>chest</code> → DenseNet121; <code>fracture</code> → YOLOv8 and the HF fracture screening classifier; <code>wound</code> → ViT.	M
FR-021	For chest scans, the system shall return at most the top 6 findings ranked by descending confidence.	M
FR-022	For fracture scans, the system shall return bounding boxes only for positive fracture detections. <code>Not_Fracture</code> detector boxes shall be ignored, because absence of a detection does not prove the scan is normal.	M
FR-023	The system shall pass all findings, the scan type and the user's notes to the LLM, which shall return an urgency rating, a synthesis paragraph, recommended actions and a suggested specialist.	M
FR-024	If LLM synthesis fails, the system shall still return a complete result using a fallback synthesis with <code>urgency: "medium"</code> . The analysis shall not fail.	M
FR-025	If image upload to storage fails, the system shall still return the analysis result. The upload shall be non-blocking.	M
FR-025a	The system shall record and return per-request diagnostics: models run, model errors, and processing time in milliseconds.	S

Urgency vocabulary (closed set): `critical` , `high` , `medium` , `low` , `clear` **Severity vocabulary (closed set):** `critical` , `high` , `moderate` , `low`

4.3 Module: Results & Records (FR-026 – FR-033)

ID	Requirement	Priority
FR-026	The system shall persist every completed analysis with its findings, synthesis, image URL, urgency and timestamp.	M
FR-027	The results page shall render the uploaded image with bounding-box overlays drawn at the correct scale for any viewport size.	M
FR-028	The results page shall display, per finding: name, confidence percentage, severity, originating model, anatomical region and ICD-10 code where available.	M
FR-028a	Findings shall be grouped into three confidence tiers: Primary ($\geq 65\%$), Secondary (50–65 %) and Borderline ($< 50\%$, collapsed by default).	M
FR-028b	If any finding has confidence below 60 % , the results page shall display a banner recommending radiologist review.	M
FR-028c	The results page shall provide image controls: toggle AI finding boxes, toggle heatmap overlay, toggle labels, and zoom between 0.5× and 2.5× with reset.	S
FR-028d	The results page shall display a confidence summary table listing every finding with its model, confidence and severity.	S
FR-029	The system shall list the signed-in user's scan history in reverse-chronological order.	M
FR-030	The system shall allow filtering scan history by scan type.	S
FR-031	The system shall generate a formatted PDF report for any of the user's scans, streamed as a download.	M
FR-032	The system shall export the raw findings of any scan as JSON .	S
FR-033	The system shall allow a user to delete their own scan; both the record and its stored image shall be removed.	M

4.4 Module: Dashboard Analytics (FR-034 – FR-037)

ID	Requirement	Priority
FR-034	The dashboard shall display total scan count, count of critical + high urgency findings, average model confidence, and average report generation time.	M
FR-035	The dashboard shall display a distribution of finding names by frequency.	S
FR-036	The dashboard shall display the user's most recent scans, each linking to its full result.	S
FR-036a	The dashboard shall display per-model average confidence (DenseNet121, YOLOv8, ViT).	S
FR-037	All dashboard figures shall be computed from only the signed-in user's own scans.	M

4.5 Module: Health Chatbot (FR-038 – FR-043a)

ID	Requirement	Priority
FR-038	The system shall provide a conversational health Q&A chatbot.	M

ID	Requirement	Priority
FR-039	The chatbot shall support English (en) and Urdu (ur) , selected per message, using a language-specific system prompt. Urdu input shall render right-to-left.	M
FR-040	The chatbot shall include the previous 10 messages of the session as context in each request.	M
FR-041	Every chatbot reply shall be constrained by a system prompt that forbids definitive diagnosis and directs users to a qualified professional.	M
FR-042	The system shall extract <code>DOCTOR_TYPE</code> and <code>HOME_REMEDIES</code> from the model reply and return them as structured fields for display in styled cards.	S
FR-043	Chat sessions and messages shall be persisted and restored on the user's next visit; a new session shall be created automatically when <code>session_id</code> is null.	M
FR-043a	The chatbot shall accept voice input via the browser SpeechRecognition API, using locale <code>en-US</code> or <code>ur-PK</code> to match the selected language.	C

4.6 Module: Diet Planner (FR-044 – FR-048)

ID	Requirement	Priority
FR-044	The system shall generate a 7-day meal plan containing breakfast, lunch and dinner for each day, plus optional snacks.	M
FR-045	The user shall be able to specify a medical condition, dietary preference, restrictions and health goals.	M
FR-046	The system shall apply condition-specific safety rules — DASH guidance for hypertension, low-glycaemic-index guidance for diabetes, and a renal-dietitian referral note for kidney disease.	M
FR-047	The system shall validate the generated plan; if it does not contain exactly 7 complete days, a validated hard-coded fallback plan shall be returned instead of a malformed one.	M
FR-048	Each meal shall include a name, a description and, where available, calories and key nutrients.	S

4.7 Module: Clinic Locator (FR-049 – FR-052)

ID	Requirement	Priority
FR-049	The system shall accept latitude, longitude and a search radius in kilometres (0.5 – 25 km , default 5 km).	M
FR-050	The system shall query the OpenStreetMap Overpass API for facilities tagged <code>hospital</code> , <code>clinic</code> , <code>doctors</code> , <code>pharmacy</code> or <code>health_centre</code> , and shall try up to three mirror endpoints before failing.	M
FR-051	The system shall compute distance using the Haversine formula and return results sorted by ascending distance.	M
FR-052	Each result shall include facility name, type, address, coordinates, distance in km and a Google Maps link.	M

ID	Requirement	Priority
FR-052a	If the browser denies geolocation permission, the UI shall display an explanatory message rather than failing silently.	S

4.8 Module: Settings (FR-053 – FR-056)

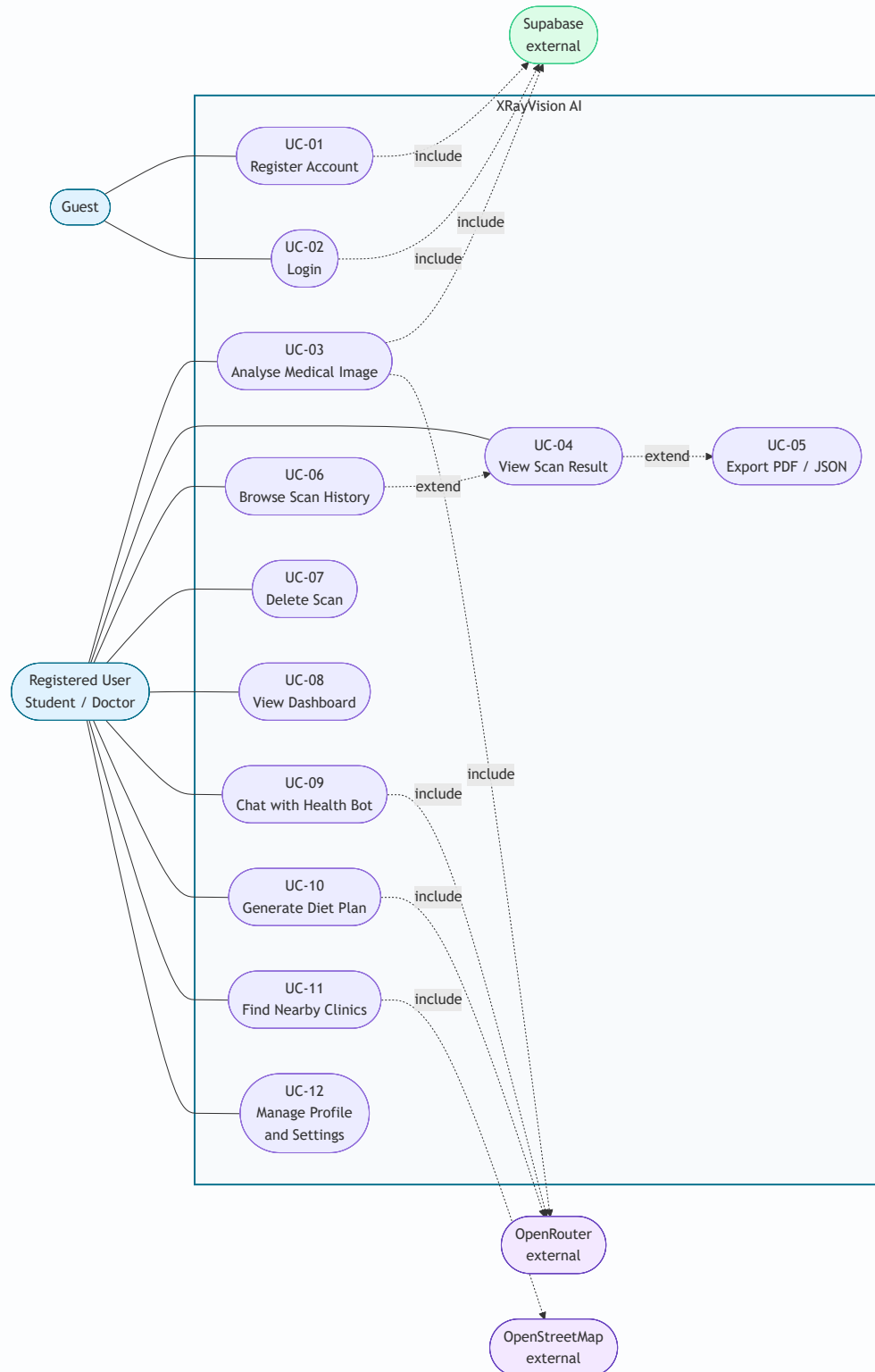
ID	Requirement	Priority
FR-053	The user shall be able to select a theme: light, dark or system.	S
FR-054	The user shall be able to toggle display of AI detection boxes on result images.	S
FR-055	The user shall be able to toggle the confidence heatmap overlay and email notifications for critical findings.	C
FR-056	Preferences shall be persisted server-side in the <code>profiles.settings</code> JSONB column.	S

4.9 Module: Cross-Cutting (FR-057 – FR-060)

ID	Requirement	Priority
FR-057	The system shall expose an unauthenticated <code>GET /health</code> endpoint returning <code>{"status": "ok"}</code> for platform health checks.	M
FR-058	The system shall expose interactive API documentation at <code>/docs</code> (Swagger UI) and <code>/redoc</code> (ReDoc).	S
FR-059	The backend shall begin accepting requests immediately at startup, with model loading performed in a background thread.	M
FR-060	When <code>DISABLE_PRELOAD=true</code> , models shall load lazily on first use so auth, chat and diet remain available on memory-constrained hosts.	M

5. Use Case Model

5.1 Use Case Diagram — System Overview



5.2 Use Case Specifications

UC-01 — Register Account

Field	Detail
Actor	Guest
Goal	Create an account and gain access to the platform
Pre-conditions	User is not authenticated; email is not already registered
Post-conditions	<code>auth.users</code> and <code>profiles</code> rows created; JWT issued; user lands on Dashboard
Trigger	User submits the registration form at <code>/auth/register</code>

Main flow

1. User opens `/auth/register`.
2. User enters full name, email, password and password confirmation.
3. System validates that both passwords match and the password is ≥ 8 characters.
4. System calls `POST /auth/register`.
5. Backend creates the user in Supabase Auth.
6. A database trigger inserts the corresponding `profiles` row.
7. Backend returns a JWT and the user profile.
8. Frontend stores the token and redirects to `/dashboard`.

Alternative and exception flows

- **A1 — Passwords do not match:** inline error at step 3; return to step 2.
- **A2 — Password too short:** "at least 8 characters" shown; return to step 2.
- **E1 — Email already registered:** backend returns 400; message displayed; return to step 2.
- **E2 — Supabase unreachable:** backend returns 500; a retry message is displayed.

UC-02 — Login

Field	Detail
Actor	Guest (returning user)
Goal	Authenticate and resume access
Pre-conditions	Account exists
Post-conditions	JWT stored in browser local storage; user on Dashboard

Main flow

1. User opens `/auth/login` .
2. User enters email and password, or clicks **Try demo account**.
3. System calls `POST /auth/login` .
4. Backend verifies credentials against Supabase Auth.
5. Backend returns JWT plus profile; frontend stores both and redirects to `/dashboard` .

Exception flows

- **E1 — Invalid credentials:** HTTP 401; "Invalid email or password"; return to step 2.
- **E2 — Demo login fails:** "Demo login failed. Please try again."

UC-03 — Analyse Medical Image (*primary use case*)

Field	Detail
Actor	Registered User
Secondary actors	OpenRouter (LLM), Supabase (persistence)
Goal	Obtain an AI-generated diagnostic report for a medical image
Pre-conditions	User is authenticated; a valid image file is available
Post-conditions	A <code>scans</code> row exists; the image is stored; the results page is displayed
Frequency	Highest-traffic use case

Main flow

1. User navigates to `/analyze` .
2. **Step 1** — User drags-and-drops or selects an image file.
3. **Step 2** — User selects an analysis route: Chest pathology, Fracture detection, or External wound.
4. **Step 3** — User optionally enters a patient/session label and notes for the AI agent.
5. User clicks **Analyze image**. A full-screen processing overlay appears.
6. System sends `POST /analyze` as `multipart/form-data` with the Bearer token.
7. Backend validates `scan_type` , then validates the file (non-empty, 100 KB – 20 MB, $\geq 200\times200$ px, decodable).
8. Backend generates a scan UUID and starts a timer.
9. Backend preprocesses the image (CLAHE + resize + normalise).
10. Backend runs the routed model(s) in a thread pool.
11. Backend sends findings, scan type and notes to the LLM for synthesis.
12. Backend uploads the image to Supabase Storage.
13. Backend inserts the complete scan record into PostgreSQL.
14. Backend returns a `ScanResult` object.

15. Frontend navigates to `/results/{scanId}` and renders the image, overlays, tiered finding cards, urgency badge and synthesis.

Alternative flows

- **A1 — No findings above threshold:** empty findings list returned with urgency `clear` ; the results page shows a "no significant findings" state.
- **A2 — DICOM file:** step 7 skips the PIL dimension check; `pydicom` handles decoding at step 9.

Exception flows

- **E1 — Empty file:** HTTP 400 "Empty file uploaded."
- **E2 — File fails quality validation:** HTTP 422 with the specific reason (too small / too large / too low resolution / undecodable).
- **E3 — Invalid scan type:** HTTP 400.
- **E4 — Not authenticated:** HTTP 401; frontend redirects to `/auth/login` .
- **E5 — Model inference fails:** HTTP 500 with the model error message; the "Analysis failed" screen is shown.
- **E6 — LLM synthesis fails:** *not an error*. Fallback synthesis used (FR-024); flow continues at step 12.
- **E7 — Image upload fails:** *not an error*. `image_url` is empty (FR-025); flow continues at step 13.
- **E8 — Rate limit exceeded:** HTTP 429 "Rate limit exceeded. Please slow down."

UC-05 — Export PDF / JSON

Field	Detail
Actor	Registered User
Pre-conditions	The scan exists and belongs to the requesting user
Post-conditions	A file is downloaded to the user's device

Main flow

1. User opens a scan result or a history row.
2. User clicks **Download PDF Report** or **Export JSON**.
3. System calls `GET /scans/{id}/report.pdf` or `GET /scans/{id}/export.json` .
4. Backend verifies ownership, renders the document and streams it.
5. The browser downloads the file.

Exception flow

- **E1 — Scan not found or not owned:** HTTP 404.

UC-09 — Chat with Health Bot

Field	Detail
Actor	Registered User
Pre-conditions	User is authenticated
Post-conditions	Message pair persisted to <code>chat_messages</code> under a <code>chat_sessions</code> row

Main flow

1. User opens `/chat`.
2. System loads existing sessions via `GET /chat/sessions`.
3. User selects a language (**EN** or **اردو**) and types a question, or dictates it using the microphone.
4. System calls `POST /chat` with `message`, `session_id` (null for a new session) and `language`.
5. Backend loads the last 10 messages of the session as context.
6. Backend applies the language-specific system prompt and calls the LLM.
7. Backend parses `DOCTOR_TYPE` and `HOME_REMEDIES` out of the reply.
8. Backend persists both messages and returns the structured reply.
9. Frontend appends the reply and renders the doctor-type and remedies cards.

Exception flows

- **E1 — Rate limit exceeded (20/min):** HTTP 429.
- **E2 — LLM unavailable:** an error message is shown; the user's text is not lost from the input.
- **E3 — Microphone permission denied:** voice button is disabled; text input remains available.

UC-11 — Find Nearby Clinics

Field	Detail
Actor	Registered User
Pre-conditions	Browser geolocation permission granted

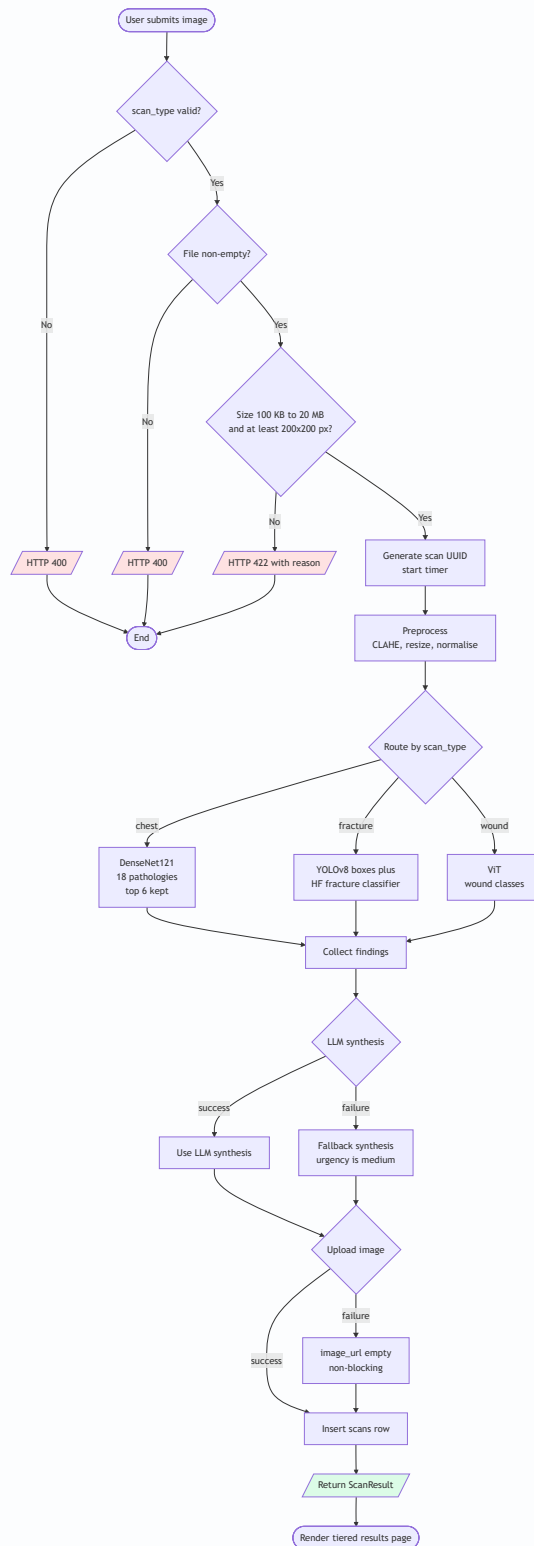
Main flow

1. User opens `/clinics`.
2. User sets the search radius with the slider (1–20 km in the UI; default 5 km).
3. User clicks **Use My Location**; the browser requests geolocation permission.
4. On grant, the frontend calls `GET /clinics?lat=..&lon=..&radius_km=..`.
5. Backend queries the Overpass API, computes Haversine distances and returns facilities sorted by distance.
6. Frontend renders name, type, address, distance and a Maps link per result.

Exception flows

- **E1 — Permission denied:** an explanatory message is shown; the API is not called.
- **E2 — No facilities within radius:** an empty-state message suggests a larger radius.
- **E3 — All Overpass mirrors unreachable:** an error message is shown and the user may retry.

5.3 Activity Diagram — Analysis Pipeline



6. Non-Functional Requirements

6.1 Performance Requirements

ID	Requirement	Target
NFR-P1	End-to-end analysis time (upload → rendered result) on a warm backend	≤ 15 s typical; 8–15 s observed
NFR-P2	Backend shall accept HTTP requests after startup without waiting for model loading	≤ 5 s to first accepted request
NFR-P3	Non-AI endpoints (<code>/scans</code> , <code>/stats</code> , <code>/auth/me</code>) response time	≤ 1 s at p95
NFR-P4	Chatbot reply latency	≤ 10 s typical
NFR-P5	Model inference shall run in a thread pool so the event loop is never blocked	<code>asyncio.to_thread</code> required
NFR-P6	Maximum accepted upload size	20 MB
NFR-P7	Concurrent analyses supported per instance	≥ 3 without failure (the rate limiter caps sustained load)
NFR-P8	Clinic search response time	≤ 30 s (Overpass timeout), typically ≤ 5 s

Note: On free-tier hosts a cold start can add 30–60 s while model weights load. This is an accepted consequence of constraint **C-1**.

6.2 Safety Requirements

ID	Requirement
NFR-S1	Every AI-generated output shall carry a visible disclaimer stating the system is for educational purposes only and is not a certified medical device.
NFR-S2	The chatbot system prompt shall prohibit definitive diagnosis and shall direct the user to a qualified medical professional.
NFR-S3	The diet planner shall apply condition-specific clinical guardrails and shall never emit an unvalidated plan.
NFR-S4	Fracture reporting shall not claim a scan is normal on the basis of a <code>Not_Fracture</code> detector box — absence of a detection is not evidence of absence of pathology.

ID	Requirement
NFR-S5	Confidence scores shall always be displayed alongside findings so the user can judge reliability.
NFR-S6	Urgency ratings shall use a fixed, documented five-level vocabulary so severity is never ambiguous.
NFR-S7	Low-confidence results (< 60 %) shall trigger an explicit recommendation for radiologist review.

6.3 Security Requirements

ID	Requirement
NFR-SEC1	Passwords shall never be stored in plaintext; hashing is delegated to Supabase Auth.
NFR-SEC2	JWTs shall be signed with HS256 using a secret of at least 32 characters. The server shall refuse to operate with the default placeholder secret.
NFR-SEC3	Row Level Security shall be enabled on <code>profiles</code> , <code>scans</code> , <code>chat_sessions</code> and <code>chat_messages</code> , restricting <code>SELECT</code> to <code>auth.uid()</code> -owned rows.
NFR-SEC4	Every scan, chat and stats endpoint shall verify ownership server-side and shall not rely on client-supplied user identifiers.
NFR-SEC5	Rate limiting shall be enforced: 200 req/min globally, 10 req/min on <code>/analyze</code> , 20 req/min on <code>/chat</code> , 10 req/min on <code>/diet</code> , keyed by client IP.
NFR-SEC6	CORS shall be restricted to configured origins in production.
NFR-SEC7	All request bodies shall be validated by Pydantic schemas before any processing.
NFR-SEC8	Secrets (API keys, service-role keys, JWT secret) shall be supplied via environment variables and shall never be committed to source control.
NFR-SEC9	Uploaded images shall be stored under a path keyed by the opaque user UUID; no patient-identifying filename shall be required.

6.4 Software Quality Attributes

Attribute	ID	Requirement
Availability	NFR-Q1	The system shall degrade gracefully: failure of the LLM, of storage, or of any single model shall not prevent a result from being returned.
Reliability	NFR-Q2	Every external call shall be wrapped in error handling; per-model failures shall be recorded in <code>model_results.model_errors</code> rather than aborting the request.
Usability	NFR-Q3	A first-time user shall be able to complete an analysis within 3 interactions from the dashboard, with no training.

Attribute	ID	Requirement
Usability	NFR-Q4	Error messages shall state the cause and the corrective action, e.g. "File is too large (24.3 MB). Maximum allowed size is 20 MB."
Maintainability	NFR-Q5	Backend code shall be organised as routers (HTTP), services (business/ML logic) and utils (persistence), with no ML logic inside a router.
Maintainability	NFR-Q6	All frontend API calls shall pass through the single client module <code>src/lib/api.ts</code> ; no component shall call <code>fetch</code> directly.
Portability	NFR-Q7	The backend shall be containerised and runnable on any Docker host with a single <code>docker run</code> plus an <code>env</code> file.
Portability	NFR-Q8	Configuration shall be environment-driven (Pydantic Settings); no host-specific value shall be hard-coded.
Interoperability	NFR-Q9	Findings shall carry ICD-10 codes where a mapping exists, so results relate to standard clinical coding.
Testability	NFR-Q10	Every endpoint shall be exercisable through the auto-generated Swagger UI at <code>/docs</code> .
Accessibility	NFR-Q11	Interactive controls shall be keyboard-reachable, carry <code>aria-label</code> s where iconographic, and urgency shall not be conveyed by colour alone.

6.5 Business Rules

ID	Rule
BR-1	A user may access only their own scans, chat sessions and statistics.
BR-2	A chest analysis returns at most 6 findings.
BR-3	A diet plan is always exactly 7 days.
BR-4	A JWT is valid for 24 hours; after expiry the user must sign in again.
BR-5	Deleting a scan deletes both the database record and the stored image.
BR-6	Deleting a user cascades to their profile, scans, chat sessions and chat messages.
BR-7	A finding is reported only if its confidence meets <code>CONFIDENCE_THRESHOLD</code> (default 0.40).

7. Other Requirements

7.1 Legal and Regulatory

ID	Requirement
LR-1	The system shall state, on the landing page, the results page and in the README, that it is an educational tool and not a certified medical device.
LR-2	The system shall not be represented as suitable for clinical diagnosis or treatment decisions.
LR-3	The software is distributed under the MIT License .
LR-4	Third-party models and libraries shall be attributed in the project documentation (TorchXRayVision, Ultralytics, the Hugging Face model authors, OpenRouter, Supabase, TanStack, Shadcn/ui, OpenStreetMap contributors).

7.2 Data Retention

ID	Requirement
DR-1	Scan records and images shall persist until the owning user deletes them or deletes their account.
DR-2	Deleting a user account shall cascade-delete all associated rows (<code>ON DELETE CASCADE</code> on every foreign key to <code>profiles</code>).

8. Appendices

8.1 Appendix A — Requirements Traceability Matrix

Requirement	Design element (SDD)	Test cases
FR-001 – FR-010a	<code>routers/auth.py</code> , <code>services/auth_service.py</code>	TC-AUTH-01 ... TC-AUTH-11
FR-011 – FR-025a	<code>routers/analyze.py</code> , <code>services/image_preprocess.py</code> , <code>services/*_model.py</code>	TC-ANLZ-01 ... TC-ANLZ-16
FR-026 – FR-033	<code>routers/scans.py</code> , <code>routes/results.\$scanId.tsx</code> , <code>routes/history.tsx</code>	TC-SCAN-01 ... TC-SCAN-12
FR-034 – FR-037	<code>routers/stats.py</code> , <code>routes/dashboard.tsx</code>	TC-DASH-01 ... TC-DASH-05
FR-038 – FR-043a	<code>routers/chat.py</code> , <code>services/chatbot_service.py</code>	TC-CHAT-01 ... TC-CHAT-08
FR-044 – FR-048	<code>routers/diet.py</code> , <code>services/diet_service.py</code>	TC-DIET-01 ... TC-DIET-06
FR-049 – FR-052a	<code>routers/clinics.py</code> , <code>routes/clinics.tsx</code>	TC-CLIN-01 ... TC-CLIN-06

Requirement	Design element (SDD)	Test cases
FR-053 – FR-056	routers/auth.py (PATCH /auth/settings), routes/settings.tsx	TC-SET-01 ... TC-SET-04
FR-057 – FR-060	app/main.py	TC-SYS-01 ... TC-SYS-05
NFR-SEC1 – SEC9	services/auth_service.py , supabase_schema.sql , main.py	TC-SEC-01 ... TC-SEC-09
NFR-P1 – P8	Whole system	TC-PERF-01 ... TC-PERF-05

Full test specifications: [docs/03-Test-Cases.md](#) .

8.2 Appendix B — Chest Pathology Classes (DenseNet121)

Atelectasis · Cardiomegaly · Consolidation · Edema · Effusion · Emphysema · Fibrosis · Hernia · Infiltration · Mass · Nodule · Pleural Thickening · Pneumonia · Pneumothorax · Enlarged Cardiomedastinum · Lung Opacity · Lung Lesion · Fracture — **18 classes**

8.3 Appendix C — Configuration Parameters

Variable	Default	Purpose
SUPABASE_URL	—	Supabase project URL
SUPABASE_KEY	—	Service-role key (server-side only)
SUPABASE_ANON_KEY	—	Public anon key
OPENROUTER_API_KEY	—	LLM API key
OPENROUTER_MODEL	z-ai/glm-4.5-air:free	LLM model identifier
OPENROUTER_TIMEOUT_SECONDS	60	LLM request timeout
HF_TOKEN	—	Hugging Face read token for model download
JWT_SECRET	—	HS256 signing secret (≥ 32 chars)
JWT_ALGORITHM	HS256	Token signing algorithm
JWT_EXPIRY_HOURS	24	Token lifetime
YOLO_WEIGHTS_PATH	models/fracture_yolov8.pt	Fracture detector weights
ALLOW_GENERIC_YOLO_WEIGHTS	false	Guard against running a non-medical YOLO checkpoint
FRACTURE_CLASSIFIER_ENABLED	true	Enable image-level fracture screening
FRACTURE_CLASSIFIER_MODEL_NAME	prithivMLmods/Bone-Fracture-Detection	HF fracture classifier
WOUND_MODEL_NAME	PayamFard123/dermaintel-wound-classifier	HF ViT wound classifier
CONFIDENCE_THRESHOLD	0.40	Minimum confidence for a finding to be reported
FRONTEND_URL	http://localhost:5173	Primary CORS origin

Variable	Default	Purpose
ALLOWED_ORIGINS	—	Additional comma-separated production origins
DISABLE_PRELOAD	false	Skip model preloading on memory-constrained hosts

8.4 Appendix D — Document Revision History

Version	Date	Author	Description
1.0	August 2026	Muhammad Ali Raza, Hamza Afzal	Initial complete SRS

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